

How syntactic complexity indices predict Chinese L2 writing quality: An analysis of unified dependency syntactically-annotated corpus

Yuxin Hao^{a,1}, Xuelin Wang^{a,b,2}, Shuai Bin^{c,3}, Qihao Yang^{d,4}, Haitao Liu^{e,*}

^a Institute of Chinese Language and Culture Education, Huaqiao University, Xiamen, China

^b College of Chinese Language and Culture, Jinan University, Guangzhou, China

^c School of Chinese as a Second Language, Peking University, Beijing, China

^d School of Computer Science, South China Normal University, Guangzhou, China

^e Department of Linguistics, Zhejiang University, Hangzhou, China

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ABSTRACT

Previous syntactic complexity (SC) research on L2 Chinese has overlooked a range of Chinese-specific structures and fine-grained indices. This study, utilizing a syntactically annotated Chinese L2 writing corpus, simultaneously employs both large-grained and fine-grained syntactic complexity indices to investigate the relationship between syntactic complexity and writing quality produced by English-speaking Chinese second language (ECSL) learners from macro and micro perspectives. Our findings reveal the following: (a) at a large-grained level of analysis using syntactic complexity indices, the generic syntactic complexity index (GSC index) number of T-units per sentence and the Chinese-specific syntactic complexity index (CSC index) number of Clauses per topic chain unit account for 14.5% of the total variance in writing scores among ECSL learners; (b) the syntactic diversity model alone accounts for 24.7% of the variance in Chinese writing scores among ECSL learners; (c) the stepwise regression analysis model, which integrates fine-grained SC indices extracted from the syntactically annotated corpus, explains 43.7% of the

Abbreviations: ECSL, English-speaking Chinese second language; SC, syntactic complexity; GSC, the generic syntactic complexity index; CSC, the Chinese-specific syntactic complexity index; NL, native language; MDD, mean dependency distance; DD, Dependency distance; MSTTRDR-50, mean 50-segmental TTR of dependency relations; TTCU, Terminal Topic-Comment Unit; TCMDD, Topic-Comment-Unit dependency distance; S, Main governor; Subj, Subject; Obj, Object; obj2, Indirect object; subobj, Subobject; soc, Subject Complement; pobj, Prepositional Object; fc, Postpositional Complements; comp, Predicate complement; dec, Complement of usde 'de 的' (the conjunction of the attribute and the head); dic, Complement of usdi 'di 地' (the conjunction of the adverbials and the head); dfc, Complement of usdf 'de 得' (the conjunction of the predicate complements and the head); Baobj, Object of the preposition 'bǎi 把'; Plc, Plural complement; Oc, Ordinal complement; Qc, Complement of classifier; Beis, Construction of Pbei 'bei 被'; Sentobj, Sentential object; Obja, Auxiliary verb; Adva, Adverbial; Va, Verb adjunct; Atr, Attributer; Top, Topic; Coor, Coordinating adjunct; Epa, Epithet; Ma, Numeral adjunct; Ta, Aspect adjunct; Esa, Adjunct of sentence end; Ina, insertion; Cr, Clause adjunct; Csr, Correlative adjunct; Auxr, Particle adjunct.

* Corresponding author.

E-mail address: htliu@163.com (H. Liu).

¹ Jimei road 668#, Jimei District, Xiamen, China. 361021

² Shougouling road 377#, Tianhe District, Guangzhou, China. 510610

³ Yiheyuan Road 5#, Haidian District, Beijing, China. 100871

⁴ West of Zhongshan Avenue 55#, Tianhe District, Guangzhou, China. 510631

⁵ Yuhangtang Rd 866#, Hangzhou, China. 310058

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variance in Chinese writing quality. This model incorporates CSC indices such as average ratio of dependency types per 30 dependency segments, the ratio of adjuncts to sentence end, the ratio of predicate complements, the ratio of numeral adjuncts, the mean length of Topic-Comment-Unit dependency distance, as well as GSC indices like the ratio of main governors, the ratio of attributers, the ratio of coordinating adjuncts, and the ratio of sentential objects. These findings highlight the valuable insights that syntactically annotated fine-grained SC indices offer regarding the writing characteristics of ECSL learners.

1. Introduction

Interlanguage is the linguistic system produced by second language learners (hereafter L2 learners) and constitutes an independent, progressively developing and dynamic language system that possesses linguistic features of both the native language and the target language (Selinker, 1972). Syntax stands as one of the fundamental pillars of human language systems, with syntactic competence playing a pivotal role in the second language acquisition process. The search for indices capable of assessing the development of syntactic competence among interlanguage has emerged as one of the paramount concerns in the realm of second language acquisition. Syntactic complexity (hereafter SC) encompasses both the sophistication of syntactic elaboration and the diversity or range of syntactic forms generated by a speaker or writer (Lu, 2011; Ortega, 2003). Syntactic complexity serves not only as a vital tool for assessing linguistic proficiency, but also as a critical component in examining language performance and language development (Kuiken et al., 2019; Lei et al., 2023; Norris & Ortega, 2009; Sánchez & Sunesson, 2023). The assumption regarding complexity in L2 learner writings suggests that texts with a higher level of syntactic complexity exhibit more fundamental and intricate syntactic structures. Conversely, texts with lower syntactic complexity feature only a limited number of basic syntactic structures (Zhang & Ouyang, 2023). To a certain extent, the advancement in syntactic complexity signifies that language learners are able to produce a wide range of syntactic patterns and provide detailed elaborations on language elements. Durrant :1) et al. (2021) conceptualize writing development as occurring along two primary axes: time and quality. In the current study, we regard the development of writing as an enhancement in quality, rather than as a temporal progression. According to Durrant :10) et al. (2021), “there is a strong tendency for L1 studies to focus on development over time and L2 studies to prefer development in terms of quality”. Additionally, while the archetypal study of quality evaluates the quality of specific texts (e.g., based on ratings assigned to those texts), a significant minority of L2 studies operationalize quality in terms of the writers who produced the texts (e.g., based on scores obtained in a separate proficiency test). We employ a focus on the development of writing in terms of quality dimensions. To summarize, in this paper, all references to ‘development’ conceptually pertain to the notion of writing development as an enhancement of quality. Our analysis focuses primarily on diverse scoring intervals, which are believed to reflect varying L2 writing competencies. Changes in these scoring intervals are indicative of alterations in the quality of writing.

According to Crossley and McNamara (2014), the method for assessing writing development involves scrutinizing how linguistic features in a text can forecast human ratings of essay quality. This approach is grounded on the assumption that the syntactic characteristics of texts function as pivotal indices of syntactic development. This is based on the belief that the inclusion of more sophisticated syntactic elements within an essay correlates with higher assessments of its quality. Over time, researchers have diligently endeavored to quantify syntactic complexity. Length per linguistic unit, density, and ratio represent the primary quantitative methods employed for gauging produced syntactic structures. These methods involve measuring the length of linguistic units, such as sentences, clauses, and T-units, while also assessing the density of specific syntactic structures. These structures encompass various types of subordinate structures, parallel structures, and different phrase types, such as complex noun phrases, prepositional phrases, and adjectival phrases (Durrant et al., 2021). Approximately hundreds of syntactic complexity indices have been employed to characterize L2 English syntactic complexity. These encompass both large-grained indices that assess overall syntactic complexity and fine-grained indices that target specific linguistic structures (Durrant et al., 2021).

Syntactic complexity is a multi-dimensional construct (Norris & Ortega, 2009). The evaluative methodologies employed for assessing syntactic complexity in L2 writing studies are continuously advancing, incorporating a spectrum of linguistic units into the measurement process to elucidate the comprehensive panorama of SC development (Jiang et al., 2019). Large-grained indices and fine-grained serve different purposes, and considering both provides a comprehensive understanding of syntactic complexity. Both large-grained and fine-grained indices hold different academic values. The significance of large-grained syntactic complexity indices lies in their practical appeal, which is further accentuated by their capacity to abstract over the nuanced intricacies of specific learner backgrounds and communicative contexts. This potential for abstraction implies a promise of universal application (Durrant et al., 2022). However, fine-grained indices empower researchers to identify the specific syntactic constructions that appear to evolve at each developmental level. They provide researchers with more detailed information about specific linguistic structures, offering heightened transparency and facilitating a more precise interpretation of variations in the quality of essays by L2 learners. Additionally, a burgeoning area of research explores the comparison between large-grained and fine-grained complexity measures, investigating their respective abilities to predict the quality of L2 writing (Jiang et al., 2019; Kyle & Crossley, 2017; Li & Yang, 2023; Zhang & Lu, 2022). There is a prevailing trend in current research suggesting that, in comparison to large-grained syntactic complexity indices, fine-grained ones demonstrate a superior capacity for predicting L2 writing quality. For instance, the findings from Zhang and Lu (2022) illustrate that fine-grained syntactic complexity indices elucidated a more substantial proportion of variance in both L2 English letter and essay quality compared to their large-grained counterparts. In the field of L2 syntactic complexity research in Chinese,

further investigation and comparison of the predictive roles of large-grained and fine-grained syntactic complexity indices on L2 writing quality should be advanced.

Nevertheless, the landscape of syntactic complexity research reveals three prominent gaps that need addressing. First, much of the research on syntactic complexity has focused on learners of English (DeVore & Kyle, 2023), the majority of well-established syntactic complexity indices have been developed for English as a second language. The absence of comprehensively processed and syntactically annotated corpora limits our ability to investigate the relationship between syntactic complexity and writing quality among learners engaged in studying Chinese as a second language. Many of the existing indices are incapable of precisely indicating the syntactic relationships between the governing elements and the dominated elements within interlanguage. We should focus our endeavors on delving into the annotation and extraction of more fine-grained syntactic complexity indicators, which are derived under the guidance of a comprehensive and systematic syntactic theory. To accomplish this, we can quantify various Chinese syntactic structures, building upon the currently limited array of experimental Chinese fine-grained syntactic indices. Second, there is a paucity of research examining the variation of L2 syntactic diversity. In previous research, syntactic complexity has been primarily operationalized through measures that focus on syntactic elaboration, such as the length of the production unit (e.g. T-unit, AS-unit) and the density of coordination, subordination, and phrasal elaboration. This approach includes assessing the ratio of selected embedded syntactic structures, like subclauses, which are considered to be developmentally or cognitively complex (Kuiken et al., 2019). Nonetheless, it is widely acknowledged that syntactic complexity can be exhibited and attained through two distinct approaches: syntactic elaboration and syntactic diversity (Bi & Jiang, 2019). The important role of syntactic diversity in measuring the syntactic complexity of L2 writing cannot be overlooked (Bulté & Housen, 2018; De Clercq & Housen, 2017). Third, the role of linguistic typological features in portraying syntactic complexity, as discussed by De Clercq and Housen (2017) and Yu (2021), has not been sufficiently explored. This oversight is particularly evident when considering the significant syntactic differences between languages like Chinese and English. While both languages have an SVO structure and bi-directional dependency, as noted by Liu (2010), Chinese displays unique typological features like topic precedence, covert connectives, and flexible sentence boundaries. This contrasts with the way English employs inflectional changes for tense formation. For example, in Chinese, verbs express tenses through particles such as 'zhe' (着), 'le' (了), and 'guo' (过), a method distinct from the approach in English for expressing tense. The existing research, including works by Jin (2007) and Yu (2021), primarily focuses on large-grained indices for L2 Chinese writing. The research exploring fine-grained indices to investigate Chinese L2 syntactic complexity from the perspective of linguistic typology is only the beginning (Hu et al., 2022; Lu & Wu, 2022).

Given the insights and gaps identified in previous research, this study will, on the one hand, continue to utilize some of the large-grained and language-independent syntactic complexity indices previously employed in English L2 syntactic complexity studies. It will explore which of these indices can effectively assess Chinese L2 writing quality, thereby examining their typological applicability. Meanwhile, by employing a range of syntactically annotated, fine-grained syntactic complexity indices from a multidimensional perspective and considering Chinese typological features, this study will further investigate the specific indices capable of assessing Chinese L2 writing quality. Therefore, the syntactic complexity indices used in this paper are categorized into two dimensions: large-grained and fine-grained indices. Within these categories, the indices are further divided into Language-Independent SC indices and SC indices with Chinese syntactic typological features. This multidimensional approach assists in pinpointing specific indices that precisely measure the quality of Chinese L2 writing.

2. Previous analysis of syntactic complexity in L2 writing

2.1. Measuring syntactic complexity of English L2 writing

Syntactic complexity indices were relatively homogeneous and large-grained in earlier research. For example, pioneering scholars have focused on large-grained indices, such as average sentence length, average clause length, T-unit length, and subordinate structure density, at the sentence level (Hunt, 1965; Ortega, 2003; Wolf-Quintero et al., 1998). Theoretically, longer syntactic units produced by language learners can be indicative of more complex grammatical structures. However, production length may not always accurately reflect the true level of syntactic complexity due to issues such as run-on sentences and the presence of parallel structures (Jiang et al., 2019; Ouyang et al., 2022). Large-grained syntactic complexity indices “collapse consideration of multiple structural and syntactic features into a single variable” (Biber et al., 2020). In a similar vein, these indices “are based solely on the number of elements in a syntactic form and do not take into account the relative frequency of the syntactic forms or the relationship between the syntactic forms and the lexical items with which they are used” (Kyle et al., 2021). Consequently, the interpretation of results obtained from these largely holistic indices may remain ambiguous.

Recently, with the advances in natural language processing facilitating the accurate automatic annotation of linguistic features, fine-grained indices that can reflect linguistic structure details have started to attract attention from scholars, aiming to create a more comprehensive, accurate, and balanced view of syntactic processing (Biber et al., 2020). Some researchers have initiated the utilization of fine-grained indices to scrutinize the developmental patterns of various subtypes of clauses and phrases in interlanguage (Biber et al., 2011; Jiang et al., 2021; Kyle, 2016; Kyle & Crossley, 2018; Mazgutova & Kormos, 2015). This transition seeks to offer a more comprehensive, precise, and well-rounded comprehension of syntactic processing. For instance, Biber et al. (2011) proposed an extensive array of clausal indices and phrasal syntactic complexity indices. Kyle (2016) identified 10 types of phrasal dependents (e.g., adjective modifiers) and seven phrase types (e.g., nominal subjects). Scholars have also explored the use of various types of noun phrases and phrasal dependent types in L2 learners (Jiang et al., 2019; Kyle & Crossley, 2018; Paquot, 2018; Zhang et al., 2022). These fine-grained based syntactic complexity studies facilitate our understanding of the diverse grammatical strategies employed by

language learners at varying proficiency levels in the construction of complex structures (Atak & Saricaoglu, 2021; Biber et al., 2011; Jiang et al., 2019; Norris & Ortega, 2009). Several significant studies have also explored the concept that learners typically transition from depending on coordination at novice levels to employing subordination, increased nominalization, and greater phrasal embedding at advanced stages of their learning journey (Norris & Ortega, 2009; Ortega, 2012). For example, the development of syntactic complexity in L2 English demonstrates a trajectory wherein there is a transition from an increased use of complex dependent clauses to a heightened use of complex phrasal structures as L2 learners' proficiency levels improve. It has been emphasized by academics that differing structural levels of syntactic complexity should be treated "with language specific calibration" (Yu, 2021). To recap, L2 syntactic research should cover as many indices of multidimensionality as possible and should probe the complexity of L2 syntax throughout various syntactic levels.

2.2. Measuring syntactic complexity of Chinese L2 writing

Comparatively speaking, researches on L2 Chinese syntactic complexity has only recently gained attention. In earlier research, Chinese second language acquisition researchers transfer T-unit-based indices to the L2 Chinese syntactic complexity study. It is debatable whether the T-unit based on clause subordination can be useful for studying the syntactic complexity of Chinese as a second language. Some academics have questioned the effectiveness of T-unit as a fundamental analytical syntactic unit to measure L2 Chinese syntactic complexity (Jin, 2007; Yuan, 2009; Yu, 2021). According to Jin (2007), the T-unit indice was proven invalid for Chinese syntactic complexity analysis. Jin (2007) argued that syntactic complexity indices that fit linguistic typological features would have a higher predictive value. Based on the topic-prominent typological characteristics in Chinese (Li & Thompson, 1976). Jin (2007) applied a topic chain in L2 Chinese complexity analysis and introduced the concept of the Terminal Topic-Comment Unit (TTCU) as well as empty categories within TTCUs for the analysis of Chinese syntactic complexity. Additionally, Jin (2007) measure the average length of topic chains, the number of clauses in topic chains, and the number of empty categories in each topic chain in the L2 Chinese writings. Yu (2021) propose the TC unit to recognize the topic chain as the underlying unit of Chinese complexity. This study finds that the mean length of terminable TC-unit and single TC-units per terminable TC-unit in combination prove the most effective for written Chinese syntactic complexity. Empty categories in TTCUs are reportedly added as a predictor of the holistic score in the stepwise regression model by Lu and Wu (2022). The resulting model explained 29.4% of the variance in holistic scores. Hu et al. (2022) demonstrated that three TC-unit-based indices (number of TTCUs, number of clauses per TTCU, and number of ZAs per TTCU) were not only significantly positively correlated with ratings of L2 Chinese writing quality but also effectively differentiated all three proficiency levels.

The research on L2 Chinese syntactic complexity indices is becoming increasingly fine-grained as the assessment methods for large-grained L2 Chinese syntactic indices. For instance, Lu and Wu (2022) emphasize the significance of fine-grained L2 Chinese syntactic complexity indices, with a specific focus on the importance of noun phrase complexity in assessing writing quality. They argue that this aspect of syntactic complexity surpasses the explanatory power of topic-comment unit indices. Hu et al. (2022) consider eight types of word combinations (including adjective-noun, adverb-predicate, verb-object, subject-predicate, classifier-noun, preposition--postposition, preposition-verb, and predicate-complement) into Chinese L2 syntactic complexity study, revealing the importance of predicate-related combinations in assessing L2 Chinese phraseological diversity and complexity. However, the fine-grained syntactic complexity indices employed by Hu et al. (2022) still require further refinement and comprehensiveness. Their study primarily concentrates on topic-comment unit-based and noun phrase complexity measures, and it lacks a more extensive exploration of Chinese-specific syntactic structures. Additionally, in Hu et al. (2022), syntactic relations were entirely automated without manual proofreading, and they did not conduct a regression analysis to examine the relationship between syntactic complexity indices and Chinese writing scores. This limitation hinders our ability to fully comprehend the influence of Chinese-specific syntactic structure indices on predicting writing scores. The LTP syntactic annotation system utilized by Hu et al. (2022) does not encompass a comprehensive and exhaustive set of Chinese syntactic relations. Based on our empirical examination of the LTP tagging system, we have observed that it comprises only 14 syntactic relations. In some cases, certain Chinese syntactic relations in the LTP tagging system actually encompass a variety of subtypes within them. For instance, the label "RAD" for dependency syntactic relation encompasses various syntactic structural relationships, including at least possessive structures with "的" (de), adverbial structures with "地" (de), Chinese tense structures, and predicate complements structures. Therefore, relying solely on fully automatic annotation of L2 learner corpora without any manual intervention may raise certain concerns. Therefore, it is important to acknowledge the fine-grained syntactic tagging system proposed by Liu and Huang (2006) based on dependency grammar theory, which comprehensively covers all syntactic structures in Chinese. This unified syntactic annotation system consists of 32 dependencies, encompassing complements and adjuncts (excluding punctuation markers). By employing a corpus annotated within this unified dependency syntactically-annotated corpus, various syntactic indices can be extracted to bridge the previous research gap, which lacked systematic and comprehensive measures of syntactic complexity in Chinese.

2.3. Measuring syntactic diversity of L2 writing

The measurement of syntactic diversity assesses the extent to which different syntactic structures or grammatical features are present in a text or linguistic production, indicating the degree of variation and richness in various syntactic forms such as phrases, clauses, sentences, and other syntactic elements (Bi & Jiang, 2020). Compared to syntactic complexity, there is currently a limited number of measurement indices for syntactic diversity. Park (2017) employed the number of verb-argument construction types (VAC types) and the corrected type-token ratio of VAC (VAC CTTR) as two diversity measures. Syntactic diversity is measured using the root

type-token ratio (RTTR) of each dependency relation in Paquot (2019). De Clercq and Housen (2017) introduced the syntactic diversity indice (SDI) and used it to investigate the language development of L2 learners. Furthermore, some researchers have employed Sentence Similarity (SYNSTUT) as a syntactic diversity indice (McNamara et al., 2014).

However, Bi and Jiang (2020) emphasized that existing syntactic diversity indices should be employed with caution. For instance, SYNSTUT is based on the "syntactic parse tree similarity between sentences" (McNamara et al., 2014), and it may be affected by a high frequency of grammatical errors, particularly incomplete structures in the language of learners, which can impact its automated analysis. SDI and VAC CTTR, on the other hand, focus solely on the diversity of specific syntactic structures, namely the diversity of clause types and verb-argument constructions. Paquot (2019) did not address an overall syntactic diversity indice in the study. Therefore, Bi and Jiang (2020) introduced a new indice for measuring syntactic diversity based on dependency relations using annotated corpora, termed MSTTRDR-50 (mean 50-segmental TTR of dependency relations). They found that MSTTRDR-50 could account for 32.1% of the variance in writing scores, demonstrating its effectiveness in explaining L2 writing quality. Syntactic diversity research emphasizes the diversity and flexibility of language, while syntactic complexity research focuses on the complexity of language structures. Combining syntactic complexity and syntactic diversity provides a more comprehensive understanding of language. Language serves not only as a structured means of conveying information but also encompasses rich modes of expression and styles. By simultaneously investigating both aspects, we can gain a deeper insight into the syntactic complexity and richness within the inter-language systems of L2 learners.

2.4. L2 Syntactic complexity studies based on syntactically-annotated Corpus

Interlanguage syntactic errors have minimal impact on dependency analysis (Bi & Jiang, 2020), rendering it advantageous for the more effective and consistent identification and extraction of grammatical structures in L2 writing (Kyle, 2016). The ability of dependency annotation to extract continuous syntactic relations from a corpus has led to increasing attention in L2 studies (Bi & Jiang, 2020). Dependency relation syntax places greater emphasis on the relationships between individual constituents that constitute a sentence, enabling the direct handling of word-to-word relationships in a sentence. It is better suited for constructing practical natural language processing systems capable of handling real-world languages, thus making it a valuable resource for syntactic research. Using dependency relations as the foundational syntactic unit for SC indices, Jiang et al. (2019) gathered data for eight fine-grained indices within the subordinate clauses and noun modifiers dimension in L2 English writings. Bi and Jiang (2020) introduced the corrected TTR of dependency relations to investigate the extent to which the syntactic diversity indice can predict the writing proficiency of secondary school English foreign language learners. Furthermore, several studies have examined clausal and phrasal complexity in English L2 learners using fine-grained indices at the dependency level (Jiang et al., 2019, 2021; Kyle & Crossley, 2018; Paquot, 2018, 2019). However, few studies have utilized similar methodologies in L2 Chinese syntactic complexity research. Dependency distance (DD), a crucial concept in dependency grammar, is considered an indice of syntactic complexity and language processing difficulty (Hudson, 2010; Liu, 2008). Researchers have observed a tendency to minimize dependency distance, as it reflects sentence structure difficulty and cognitive cost (Futrell et al., 2015; Temperley & Gildea, 2018; Liu et al., 2017). Previous studies have shown that mean dependency distance (MDD) effectively measures the development of syntactic complexity in English and Chinese L2 learners (Hao et al., 2021a; Li & Yan, 2020; Ouyang & Jiang, 2018; Ouyang et al., 2022). However, to date, there has been no study that has integrated dependency distance with Chinese topic chains.

While significant progress has been made in second language syntactic complexity research, it is imperative to recognize the limitations of previous investigations. First, the existing syntactic complexity indices in English are notably comprehensive. However, it is essential to investigate the applicability of these current syntactic complexity indices to the analysis of L2 Chinese. Fine-grained syntactic complexity indices have predominantly been applied to assess L2 writing in English, while research on these indices in Chinese is still in its early stages. Specifically, there is a lack of syntactic complexity indices that accurately capture the typological characteristics of Chinese, with only a limited number currently available. Furthermore, only a few fine-grained syntactic complexity indices in Chinese have been proposed, which neglect to compare the native language typological features of the Chinese L2 learners with the Chinese target language. Lu and Wu (2022) introduced Chinese typological indices without considering a comparison with the native language (Korean) of the Chinese learners in their study. The specific usage of this indice in Korean appears to have been given little consideration, but only compared it with Indo-European languages. Using Chinese-specific syntactic complexity indices without comparing the native language and the target language of Chinese may present certain challenges. The findings obtained by Lu and Ai (2015) strongly suggest that the L1 backgrounds of writers cannot be left unexamined when studying syntactic complexity in L2 writing (Ortega, 2015). Lu and Ai (2015) showed convincingly that L1-related forces, if unidentified in the designs, may contribute error to observations obtained from syntactic complexity indices. We argue that when developing specific syntactic complexity indices that incorporate typological features of the target language, it is crucial to conduct contrastive-linguistic comparisons to account for the linguistic differences between the native language and the target language. This approach is essential to determine the salient indices that are particularly pertinent in gauging the writing development process of L2 learners. Additionally, a paramount concern pertains to the fact that numerous distinct syntactic structures in Chinese remain largely unexplored. To address these research gaps, it is crucial to extract these syntactic structures from a corpus annotated using a unified dependency syntactically-annotated framework, such as dependency grammar, which can provide an effective solution to this issue. Additionally, few existing L2 studies have integrated syntactic diversity indices with syntactic complexity indices, which is necessary for a more comprehensive assessment of the syntactic proficiency of L2 learners. To comprehend the influence of syntactic diversity and syntactic complexity on L2 writing development, it is imperative to explore their roles in L2 acquisition and their interactions. Future research on Chinese L2 acquisition should encompass both syntactic diversity and complexity indices to achieve a comprehensive understanding of the linguistic

development of L2 learners.

3. Current research

The present study addresses the research gaps by developing a Chinese interlanguage syntactic annotation corpus using a combination of automatic and manual tagging methods. The corpus is grounded in dependency grammar theory (Hudson, 2010; Liu, 2008) and encompasses all syntactic structures in Chinese. The unified dependency syntactically-annotated corpus enables accurate and consistent analysis, providing valuable linguistic information. The present study explores the relationship between syntactic complexity and writing quality among English-speaking Chinese L2 learners (hereafter ECSL learners). To achieve this research goal, we have employed 10 large-grained indices based on sentences, clauses, T-units, and topic chains, as well as thirty-five fine-grained indices, including thirty-two syntactic dependency indices, two dependency distance indices, and one syntactic diversity indice. The ten large-grained syntactic complexity indices represent commonly utilized research indices in existing studies on L2 syntactic complexity in both Chinese or English. These syntactic indices can be automatically extracted by computer programs without the need for syntactic annotation and processing. The remaining thirty-five fine-grained syntactic complexity indices are derived from the dependency syntactically annotated Chinese interlanguage corpus. These fine-grained indices necessitate deep-level processing and annotation of L2 learner corpora at the syntactic level using a dependency syntactic analysis framework, followed by automated frequency or occurrence counting of these fine-grained indices through Python scripts. The indices used in this study can be broadly categorized into two main categories: large-grained SC indices and fine-grained SC indices. These categories are further divided into two subcategories of indices: generic syntactic complexity indices (GSC indices), which are commonly used across different languages and are independent of any specific language, and syntactic complexity indices with Chinese-specific features (CSC indices). We first extract specific fine-grained indices of syntactic complexity based on the Chinese syntactically annotated interlanguage corpus, and then compare these indices to large-grained SC indices such as the number and length of syntactic units to determine which type of syntactic complexity indices performs better in measuring the writing quality of ECSL learners. The specific research questions are:

RQ(1). What is the predictive power of the large-grained syntactic complexity indices on the writing qualities of ECSL learners?

RQ(2). What is the predictive power of the syntactic diversity indice on the writing qualities of ECSL learners?

RQ(3). What is the predictive power of fine-grained syntactic complexity indices extracted from a syntactically annotated corpus on the writing qualities of ECSL learners?

RQ(4). In the dimensions of both large-grained and fine-grained SC indices, which GSC and CSC indices can explain the writing quality of Chinese L2 learners?

4. Research methods

4.1. Participants and materials

The writing essays data utilized in this research were extracted from The Chinese Proficiency Test Dynamic Composition Corpus. The Chinese Proficiency Test (also known as the *Hànyǔ Shuǐpíng Kǎoshì*, hereafter HSK) is a standardized criterion-referenced test developed to measure the proficiency level of non-native speakers of standard Chinese, which is similar to TOEFL for English. The grading of the writing section of the HSK is subject to strict and standardized procedures. Evaluators are individuals who hold advanced degrees, either at the master's or doctoral level, and possess extensive experience in teaching Chinese as a second language. The evaluation process commences with the meticulous selection of representative benchmark essays, followed by rigorous training of graders according to the scoring criteria. Subsequently, two language assessment experts independently evaluate the essays using a blind grading system. After the initial grading, a relevance check is performed. In the event of inadequate concordance among evaluators, an additional evaluator is engaged to reassess the essays, ensuring the attainment of an equitable and unbiased score. Researchers have repeatedly demonstrated that the reliability coefficient of the HSK, as an authoritative language test, consistently exceeds 0.950 (Sun, 2007). Based on the available technical reports and research, the assessment records of HSK exhibit a remarkable level of uniformity both among the test items and across different genders (Peng et al., 2021). The HSK scoring rubric employs a holistic rating scale that includes analytic components, specifically content and structure, grammar, as well as Chinese character and lexical knowledge. It should be noted that the essay scores within the HSK scoring system are not entirely continuous. In the metadata, the testing organization allocates numeric scores to the writing dataset's 12 levels in 5-point increments, from 40 to 95, resulting in 12 distinct categories (Wang & Hu, 2021).

This study takes into account the potential impact of the essay topics or writing prompts on the syntactic complexity in learner texts (Alexopoulou et al., 2017; Zhang & Lu, 2022). However, a disparity was observed in the distribution of essays across different scoring segments. Upon controlling for variables such as theme, topics, and authors' native language backgrounds, the limited availability of essays in the lower score ranges, particularly those scoring 40, 45, 50, 55, and 60, became evident. Similarly, the corpus demonstrated a scarcity of essays in the higher scoring brackets, especially at the 90 and 95-point levels. To ensure uniformity across score categories, essays with scores of 40, 45, 50, 55, and 60 were grouped together, deemed to represent the lowest quality. In a similar manner, essays scoring 90 and 95 were merged, denoted as the highest quality. Essays scoring 65, 70, 75, 80, and 85 were sufficiently numerous, negating the need for merging. Ultimately, a careful selection was made of 245 HSK essays, with scores spanning from 40 to 95 points. Each scoring bracket contained 35 essays, contributing to a total of 38,935 words (punctuation excluded). The scores constitute an arithmetic sequence, with each differing by 5.

This study, informed by the methodologies of Yoon (2018) and Jiang et al. (2019), operationalizes writing proficiency in terms of

writing scores and defines the development of writing proficiency as an increase in these scores. Our decision to concentrate on writing scores rather than entire examination scores is based on the understanding that total scores encompass overall performance and not just the performance in writing tasks, as Kisselev et al. (2022a),(2022b) have noted. To assess writing proficiencies, a one-way analysis of variance (ANOVA) was used to examine if the segmented score bands could differentiate the writing ability of native English speakers. The results revealed a significant difference in writing scores among the seven score ranges of ECSL learners ($ps < .001$), indicating the meaningfulness of the segmented scores. We ensured that the genre and topic of each composition were the same across each score range. Essay topics were familiar to L2 learners, such as 'Remember a person who has the greatest influence on me' or 'The joys and frustrations in learning Chinese'. In the essays we selected, it was noted that candidates predominantly described specific events. Typically, narrative themes are centered around stories or events narrated in the past tense, as explicated by Kormos (2011) and Yoon (2021). Narrative writing demands an augmented use of personal pronouns, alongside adverbials that specify place and time, as highlighted by Yoon (2021). The multidimensional metric data for writings produced by ECSL learners, narrative texts by native Chinese speakers, and argumentative texts by native Chinese speakers were further calculated, encompassing indicators such as first, second, and third person pronouns, adverbs indicating past time, locative and temporal adverbials, aspectual particles for past tense, content words, function words, and abstract nouns. This data then underwent hierarchical cluster analysis utilizing the Euclidean distances between text vectors. The results indicate that the compositions of ECSL learners closely cluster with the narrative texts of native Chinese speakers. This clustering implies that these two categories of texts share similarities in the use of these multidimensional linguistic features. In other words, based on objective data, the compositions of ECSL learners show a greater inclination towards the style of narrative texts over argumentative texts in Chinese. To mitigate the impact of text length on syntactic complexity indices calculation, we controlled each length of essay to approximately 160 words (excluding punctuation). Table 1 provides word counts and writing scores for the corpora of ECSL learners.

4.2. Indices and Research Tools

The current study includes 10 large-grained syntactic complexity indices for two domains (length and amount). Specifically, our large-grained syntactic complexity indices include the mean length of sentence (MLS), the mean length of clauses (MLC), the mean length of T-units (MLTU), the number of T-units per sentence (T/S), and the number of clauses per sentence (C/S), which are more commonly used in English. We also incorporated topic chains, a Chinese-specific fine-grained indice, chosen for its proven suitability in mandarin Chinese based on previous studies (Hu, 2022; Lu & Wu, 2022). In this article, we use the mean length of topic chains (MLTCU), the mean length of topic chain clauses (MLCTCU), the number of clauses per topic chain unit (C/TCU), the number of topic chain units (NTCU) and the number of empty categories in topic chain units (NTCE). The specific definitions can be found in Jin (2007), Yu (2021) and Lu and Wu (2022). These 10 large-grained syntactic complexity indices can all be extracted without the need for dependency syntactic annotated corpora. A summary of the large-grained syntactic complexity indices is presented in Table 2. In general, the linguistic features assessed by the selected large-grained SC indices are shared between both Chinese and English, except for the specific linguistic feature of the "topic chain", which is unique to Chinese.

The syntactic diversity and fine-grained syntactic complexity (SC) indices utilized in this study are extracted from the dependency syntactically annotated Chinese interlanguage corpus based on dependency grammar theory. The interlanguage corpus used in this study, which follows a unified dependency syntactically-annotated corpus, encompasses all syntactic structures in Chinese, rendering it highly comprehensive for syntactic analysis. A syntactically-annotated corpus based on dependency grammar for ECSL learners has been constructed in this research. Dependency grammar views the sentence as a complete unit structured by the relationships between constituent linguistic units. These relationships involve a dominant element and a dependent element, forming directed dependencies (Liu, 2010). Dependency types facilitate the extraction of target syntactic information and continuity indices systematically and effectively. The dependency tree is a graphical representation of dependency relations, as illustrated in Fig. 1. In this figure, *Subj*, *Obj*, *Adva*, and *Pobj* above the arc denote subject-predicate, predicate-object, adverbial-predicate, and prepositional relations, respectively. The labels *r*, *p*, *n*, and *v* represent pronouns, prepositions, nouns, and verbs, respectively, while the numbers 1 to 5 indicate word order.

To convert the information from Fig. 1 into Table 3, Python scripts were employed for computational convenience.

To ensure the purity of the composition corpus, we removed word and punctuation errors using the error tagging system in the HSK dynamic composition corpus. The resulting corpus was then syntactically annotated, including word segmentation, POS tagging, and syntactic dependency annotation. Both machines and linguistic experts collaborated in the dependent syntactic annotation process. The Language Technology Platform (LTP) (Che et al., 2021) was used for first automated part of speech annotation and dependency relation tagging. Then we enhanced and extended the syntactic information through manual annotation of the initially

Table 1
A brief information of each score range.

Average score of essays (min, max)	Number of essays	Tokens
40-60	35	5553
61-65	35	5508
66-70	35	5578
71-75	35	5672
76-80	35	5565
81-85	35	5509
86-95	35	5550

Table 2

A description of large-grained complexity indices.

Dimension	Category	Label	Definition
Length of production unit	GSC indice	MLS	Mean length of sentences
		MLC	Mean length of clauses
		MLTU	Mean length of T-units
	CSC indice	MLTCU	Mean length of topic chains
Amount of production unit	GSC indice	MLCTCU	Mean length of topic chain clauses
		T/S	Number of T-units per sentence
		C/S	Number of Clause per sentence
	CSC indice	C/TCU	Number of Clauses per topic chain unit
		NTCU	Number of topic chain units
		NTCE	Number of empty categories in topic chain units

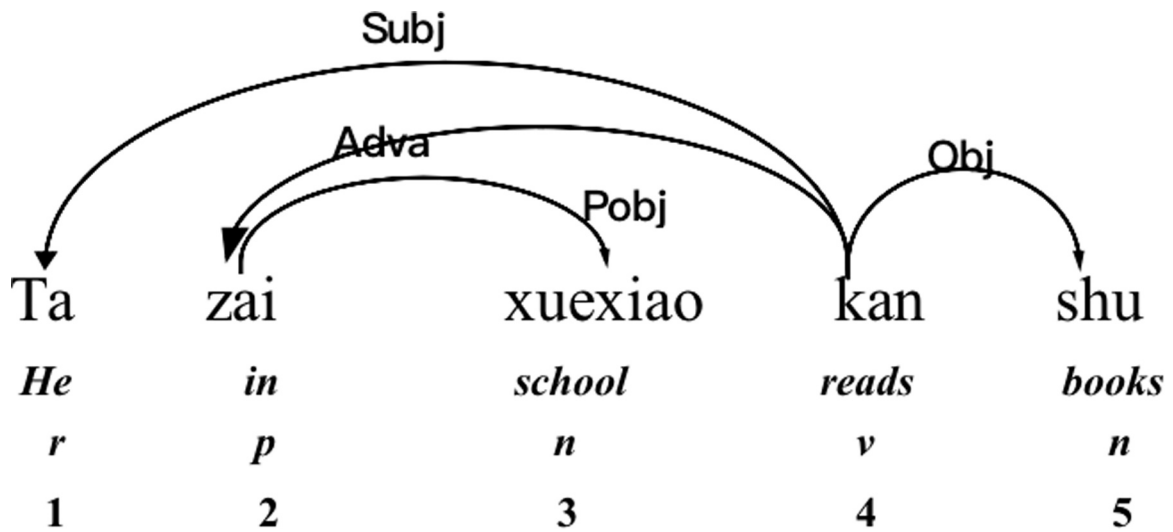
**Fig. 1.** An analysis of the dependency structure of the Chinese sentence "He reads at school".**Table 3**

Table of dependency syntactic structures for "He reads books at school".

Dependent			Governor			Dependency	DD	ADD
Number	Word	POS	Number	Word	POS			
1	Ta (He)	r	4	kàn (reads)	v	Subj	3	3
2	zài (at)	p	4	kàn (reads)	v	AdvA	2	2
3	xuéxiào (school)	n	2	zài (in)	p	Pobj	-1	1
4	kàn (reads)	v	0	/	/	S	/	/
5	shū (books)	n	4	kàn (reads)	v	Obj	-1	1

machine-labeled data using the syntactic annotation framework proposed by [Liu and Huang \(2006\)](#). Expanding upon the syntactic annotation system of [Liu and Huang \(2006\)](#), we developed thirty-five Chinese fine-grained indices that capture both syntactic complexity and diversity. Three linguistic experts were engaged in the annotation process, with each expert annotating and cross-referencing the corpus on a minimum of three occasions. In instances of disagreement, they convened discussions to reach a consensus for the ultimate sentence annotation.

To assess the evolution of overall syntactic diversity in Chinese interlanguage, this study followed the research approach proposed by [Bi and Jiang \(2020\)](#), which utilizes the corrected TTR of dependency relations. However, we adjusted the Chinese syntactic diversity indice to MSTTRDR-30 to accommodate the more than 30 types of Chinese dependency relationships present in the syntactic annotation system we employed. Therefore, this article posit that the syntactic diversity indice utilized in this article can be classified as a Chinese-specific syntactic diversity indice. In this study, MSTTRDR-30 is also considered a fine-grained syntactic complexity indice, as it is computed as a syntactic diversity indice following fine-grained syntactic annotation. The syntactic complexity measures used in this study were selected as the mean length of dependency distance (MDD), the mean length of Topic-Comment-Unit dependency distance (TCMDD), and the usage ratio of 32 syntactic dependency types in the annotated corpus. Based on the syntactic tagging corpus, we identified thirty-five fine-grained syntactic complexity indices, as presented in [Table 4](#). It is noteworthy that the

mean length of Topic-Comment-Unit dependency distance is a distinct feature of Chinese among the linguistic characteristics of MDD related indices. Among the linguistic features of the dependency relationship, *Adva*, *Atr*, *Obj*, *Subj*, *S*, *Cr*, *Coor*, *Epa*, *Pobj*, *Obj_a*, *Obj₂*, and *Sentobj* have a lot of similarities in Chinese and English. In comparison, Chinese has typological language-specific dependency syntactic relations such as *Dec*, *Ma*, *Csr*, *Qc*, *Comp*, *Fc*, *Ta*, *Soc*, *Auxr*, *Subobj*, *Oc*, *Plc*, *Va*, *Esa*, *Ina*, *Top*, *Dic*, *Dfc*, *Dec*, *Beis*, and *Baobj*. In this study, we employed a custom-developed Python script, specifically designed for this purpose, to calculate the syntactic complexity indices of ESCL learners. The accuracy of the Python program was verified by manually checking the results for 20 sample essays, achieving a 100% accuracy rate.

5. Results

Initially, Python scripts were employed to compute all syntactic complexity index data utilized in the ESCL writings, followed by conducting descriptive statistics and statistical analyses within SPSS 25.0. These procedures aimed to provide a comprehensive explanation of the extracted index information. The statistical tests encompassed Pearson correlation analysis, multicollinearity analysis, and linear stepwise regression analysis. We conducted separate correlation and regression analyses in [Sections 5.1 and 5.2](#) to examine the relationships between Large-Grained Syntactic Complexity indices and Fine-Grained Syntactic Complexity indices with writing performance. We developed three linear regression models: a multiple regression model for large-grained syntactic complexity indices, a simple regression model for the syntactic diversity index, and a comprehensive multiple regression model for fine-grained syntactic complexity indices. These models were employed to evaluate the predictive capability of large-grained syntactic complexity indices, the syntactic diversity index, and fine-grained syntactic complexity indices on the writing quality of ECSL learners. In this study, we report the predictive results of the multiple regression model for large-grained syntactic complexity indices in [Section 5.1.3](#), the simple regression model for syntactic diversity indicators in [Section 5.2.1](#), and the comprehensive multiple regression model for fine-grained syntactic complexity indicators in [Section 5.2.4](#).

5.1. Large-grained indices

5.1.1. Large-Grained GSC Index

[Table 5](#) shows the descriptive statistics and Pearson's correlations between 5 large-grained GSC indices and the overall scores.

Table 4

A description of fine-grained complexity indices based on dependency syntactically-annotated corpus.

Dimension	Category	Label	Definition
Grammatical diversity	CSC index	MSTTRDR-30	Average ratio of dependency types per 30 dependency segments
Dependency Distance	CSC index	TCMDD	The mean length of Topic-Comment-Unit dependency distance
	GSC index	MDD	The mean length of dependency distance
Dependency relationship ratio	GSC index	Adva_RATIO	The ratio of adverbials
	GSC index	Atr_RATIO	The ratio of attributers
	GSC index	Obj_RATIO	The ratio of objects
	GSC index	Subj_RATIO	The ratio of subjects
	GSC index	S_RATIO	The ratio of main governors
	GSC index	Cr_RATIO	The ratio of clause adjuncts
	GSC index	Pobj_RATIO	The ratio of prepositional objects
	GSC index	Obj _a _RATIO	The ratio of auxiliary verbs
	GSC index	Coor_RATIO	The ratio of coordinating adjuncts
	GSC index	Sentobj_RATIO	The ratio of sentential objects
	GSC index	Epa_RATIO	The ratio of epithets
	GSC index	Obj ₂ _RATIO	The ratio of indirect objects
	GSC index	Csr_RATIO	The ratio of correlative adjuncts
	CSC index	Ma_RATIO	The ratio of numeral adjuncts
	CSC index	Qc_RATIO	The ratio of complement of classifiers
	CSC index	Comp_RATIO	The ratio of predicate complements
	CSC index	Fc_RATIO	The ratio of postpositional complements
	CSC index	Ta_RATIO	The ratio of aspect adjuncts
	CSC index	Soc_RATIO	The ratio of subject Complements
	CSC index	Auxr_RATIO	The ratio of particle adjuncts
	CSC index	Subobj_RATIO	The ratio of subobjects
	CSC index	Oc_RATIO	The ratio of ordinal complements
	CSC index	Plc_RATIO	The ratio of plural complements
	CSC index	Va_RATIO	The ratio of verb adjuncts
	CSC index	Esa_RATIO	The ratio of adjuncts to sentence end
	CSC index	Ina_RATIO	The ratio of insertions
	CSC index	Top_RATIO	The ratio of topics
	CSC index	Dic_RATIO	The ratio of complements to usdi 'di 地'
	CSC index	Dfc_RATIO	The ratio of complements to usdf 'de 得'
	CSC index	Dec_RATIO	The ratio of complements to usde 'de 的'
	CSC index	Beis_RATIO	The ratio of constructions to Pbei 'bei 被'
	CSC index	Baobj_RATIO	The ratio of objects to the preposition 'ba 把'

Through Pearson's correlation analysis using SPSS, among the length-related indices, all except MLTU are significantly correlated with Chinese L2 writing scores ($ps < 0.05$). Higher writing scores are associated with longer average sentence lengths, decreased average clause lengths, increased numbers of T-units per sentence, and increased numbers of clauses per sentence.

Fig. 2 provides a visual view of the development of large-grained GSC indices. To begin, we focus on generic length SC indices. As depicted in Fig. 2(a), the improvement in Chinese essay scores among ECSL learners corresponds with a significant rise in the MLS value, from 17.641 to 21.676, signifying ECSL learners' ability to construct progressively longer sentences. Figs. 2(b) and 2(c) show that ECSL learners produce longer clauses and T-units at the 90–95 score level. Compared to the 40–60 score level, the values of MLC and MLTU show a slight but statistically insignificant decrease at the 90–95 score level. Second, we observe the development of generic amount SC indices. The data presented in Figs. 2(d) and 2(e) exhibit a discernible upward trend characterized by fluctuations in both the T/S and C/S ratios.

5.1.2. Large-grained CSC Index

Based on Table 6, the 245 analyzed essays have an average of 16.084 words per topic chain and 6.1025 words per topic chain clause. In the same dimension, the writings of participants exhibit 2.178 clauses per topic chain unit, 2.062 topic chain units, and 3.522 empty categories in topic chains. Through the correlation analysis conducted using SPSS, with the exception of MLCTCU, all other five large-grained CSC indices exhibited a significant correlation with Chinese L2 writing scores ($ps < 0.05$). Moreover, as L2 Chinese writing scores improved, ECSL learners showed an increase in the number of topic chains, their lengths, clauses per topic chain unit, and empty categories.

Fig. 3 demonstrates that as the Chinese writing scores of ECSL learners improve, there is a gradual increase in the average length of their Chinese sentence topic chains and sentence topic chain clauses. For instance, the average length of sentence topic chains rises from approximately 12.760 in the 40–60 score range to 17.814 in the 90–95 score range, while the average length of sentence topic chain clauses increases from around 5.751 in the 40–60 score range to 6.243 in the 90–95 score range. Notably, a critical juncture at the score of 65 in the development of MLTCU is observed, with a discernible disruption in its progression persisting until a score of 75. This suggests that with a score exceeding 65, there is a significant tendency among ECSL learners to extend the length of their topic chains in their essays, resulting in the formation of more complex and elongated chains. Consequently, ECSL learners are capable of producing longer and more sophisticated topic chains. Furthermore, the overall variation in the average length of sentence topic chains is greater than that of sentence topic chain clauses. Figs. 3(c), 3(d), and 3(e) demonstrate that C/TCU, NTCU, and NTCE each increase as writing scores improves, indicating that the typological features of ECSL learners gradually approximate the topic-prominent typological characteristics of Chinese.

5.1.3. Predictive Power of Large-Grained Syntactic Complexity Indices in Predicting Writing Scores

We conducted a stepwise regression analysis in SPSS to investigate the relationship between syntactic complexity indices and Chinese L2 writing scores. The results indicate that both T/S and C/TCU are included in the stepwise linear regression model. Furthermore, the estimated standard deviation error for both sets of regression models is lower than the standard deviation of the dependent variable, which is the writing scores for ECSL learners (11.795). Both T/S and C/TCU have a tolerance greater than 0.5 and a variance inflation factor (VIF) less than 2 after entering the regression model. This indicates that there is no multicollinearity in this model. This answers Question (1). As presented in Table 7, T/S and C/TCU explained 14.5% of the total variance in the writing scores of ECSL learners. The observed residuals and the scaled versions of the observed residuals have been utilized extensively to examine the validity of regression models and their assumptions (Rawlings et al., 1998). The standardized residual histogram and P-P plot of the large-grained syntactic complexity model follow a normal distribution, and the coefficient of regression model is significant ($p < .005$), confirming its successful establishment. The R^2 of the fine-grained regression model for ECSL learners obtained a medium effect size in accordance with Cohen (1988) reference system of effect sizes.

Although large-grained indices can reflect some degree of syntactic development of ECSL learners, we cannot probe the internal syntactic structure changing of their language production. In the following, we will adopt fine-grained indices from the syntactically annotated corpus as the evaluation indice to reveal a microscopic picture of the syntactic change of ECSL learners.

5.2. Fine-Grained Indice

We utilized thirty-five fine-grained syntactic complexity indices, covering syntactic variety, syntactic dependency distance (MDD and TCMDD), and 32 dependency relationship ratios, to examine the developmental patterns in the Chinese L2 writing abilities of ECSL learners. Table 8 presents descriptive statistics for the syntactically annotated corpus-based syntactic complexity fine-grained indices,

Table 5

Descriptive statistics of large-grained GSC indices and their correlation coefficients with holistic score.

Category	indice	Mean (SD)	Correlation with holistic score
GSC indice	MLS	20.190 (5.674)	.273
	MLC	7.630 (1.364)	-.133
	MLTU	9.301 (1.629)	-.079
	T/S	2.713 (0.846)	.343
	C/S	2.217 (0.671)	.312

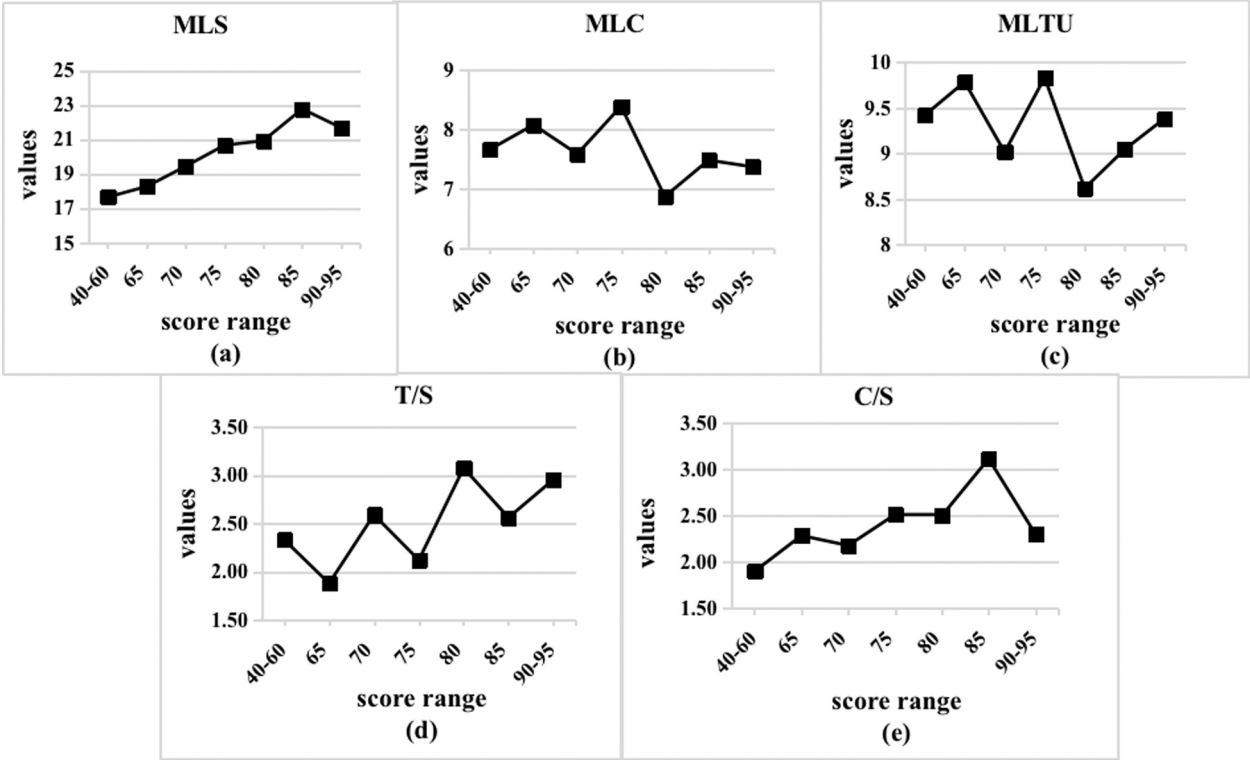


Fig. 2. Development trends of large-grained GSC indices across L2 Chinese writing scores.

Table 6

Descriptive statistics of large-grained CSC indices and their correlations with holistic score.

Category	indice	Mean (SD)	Correlation with holistic score
CSC indice	MLTCU	16.084 (8.953)	.254
	MLCTCU	6.103 (2.964)	.116
	C/TCU	2.178 (1.016)	.282
	NTCU	2.062 (1.397)	.134
	NTCE	3.522 (3.043)	.223

along with their correlations with the overall scores for ECSL learners.

5.2.1. Fine-Grained Syntactic Diversity Index

In the assessment of the writing quality of Chinese L2 learners, syntactic diversity holds an equally crucial position. Therefore, we have chosen to initially isolate the syntactic diversity indice, MSTTRDR-30, for analysis and investigate its effectiveness in predicting Chinese second language writing qualities in response to Question (2). We multiplied the value of MSTTRDR-30 by the denominator to convert frequency values into frequency counts for analysis. On average, the 245 essays written by Chinese learners have 12.672 syntactic relationships within 30 dependency relationship groups. As shown in Fig. 4, there is a significant correlation between MSTTRDR-30 and Chinese writing qualities ($p < .001$, $r = .497$). The MSTTRDR-30 indice has shown an increase from 0.402 to 0.446, suggesting that ECSL learners exhibit the capacity to employ 12.06, 12.18, 12.36, 12.45, 12.72, 13.56, and 13.38 types of syntactic relations across a range of writing scores from 40 to 95, respectively.

The regression results are shown in Table 9. The statistics demonstrate that the syntactic diversity indice MSTTRDR-30 alone accounts for 24.7% of the variance in writing performance, and is even higher than the overall explanatory validity of the large-grained syntactic complexity model, with a significant regression coefficient ($p < .05$). This answers Question 2. Presented results indicate that MSTTRDR can also be used as an effective indice of the syntactic diversity in Chinese L2 learners. This further suggests that the corrected TTR of dependency relations have good predictive power not only for L2 English writing qualities but also for the L2 Chinese qualities. According to Verspoor et al. (2017), syntactic structures in interlanguage change as L2 learners become more adept in the second language. This study illustrates that as L2 Chinese writing scores improve, ECSL learners tend to utilize a broader range of Chinese syntactic structures in their writings.

5.2.2. Fine-grained syntactic dependency distance

ECSL learners have an average dependency distance of 2.644 and an average dependency distance of 2.242 for producing topic chains. Fig. 5(a) and 5(b) demonstrate an increasing trend in both MDD and TCMDD of ECSL learners.

As the Chinese writing scores of ECSL learners increasing, the number of syntactic components between words with a dependency relation also increases. This expansion of syntactic hierarchies and higher syntactic complexity suggests that ECSL learners are more likely to produce sentences with sophisticated syntactic structures in their essays. The rise in dependency distance suggests that ECSL learners are enhancing their proficiency in employing diverse embedded syntactic structures within their sentences, indicating an improvement in their ability to incorporate a broader array of such structures. This aligns with similar findings in the research conducted by Ouyang and Jiang (2018) and Li and Yan (2020) on English as a second language acquisition. This suggests that MDD, as an effective indice for assessing interlanguage syntactic development, holds a degree of general applicability. The positive correlation between MSTTRDR-30 values and writing scores, as depicted in Fig. 4, provides further support for this observation. These findings indicate that ECSL learners are progressively acquiring the ability to construct a broader range of diverse and complex syntactic relationships. In the next section, we will explore specific dependency relationship ratios and provide examples to demonstrate the variation in writing quality among ECSL learners, illustrating how different dependency arrangements contribute to this analytical process.

5.2.3. Fine-Grained Dependency Relationship Ratio

Considering the constraints of journal publication space, this section will primarily concentrate on representative and notably diverse fine-grained syntactic relationship indices. We will commence with an examination of the fine-grained GSC dependency relationship indices.

Fig. 6(a) reveals a declining trend in the use of Subj_RATIO in the writings of ECSL learners. This decline implies that ECSL learners are becoming proficient in constructing sentences without requiring explicit subjects. This phenomenon aligns with the topic-prominent linguistic typological feature of Chinese, which permits subject omission or implication. In contrast, English is a subject-prominent language where the subject holds a prominent role (Li & Thompson, 1976). Generally speaking, in English, except for imperative sentences and informal registers, a sentence must have a subject, and there exist non-personal, functional subjects such as "It" or expletive "There" that have weaker semantic meaning but serve as subject in English syntax. ECSL learners have the highest usage rate of Subj within the 40–60 point range. This observation indicates that ECSL learners exhibit a significant transference of the subject-prominence feature from their native language (English) into their lower-quality Chinese writings, consequently resulting in a higher frequency of explicit subjects. This phenomenon aligns with the findings presented in the research conducted by Jin (1994) and Yuan (1995). With the enhancement of writing qualities among ECSL learners, they increasingly utilize null subjects in their Chinese writings, particularly when compared to their earlier writings of lower quality. It is suggested by this observation that learners of ECSL

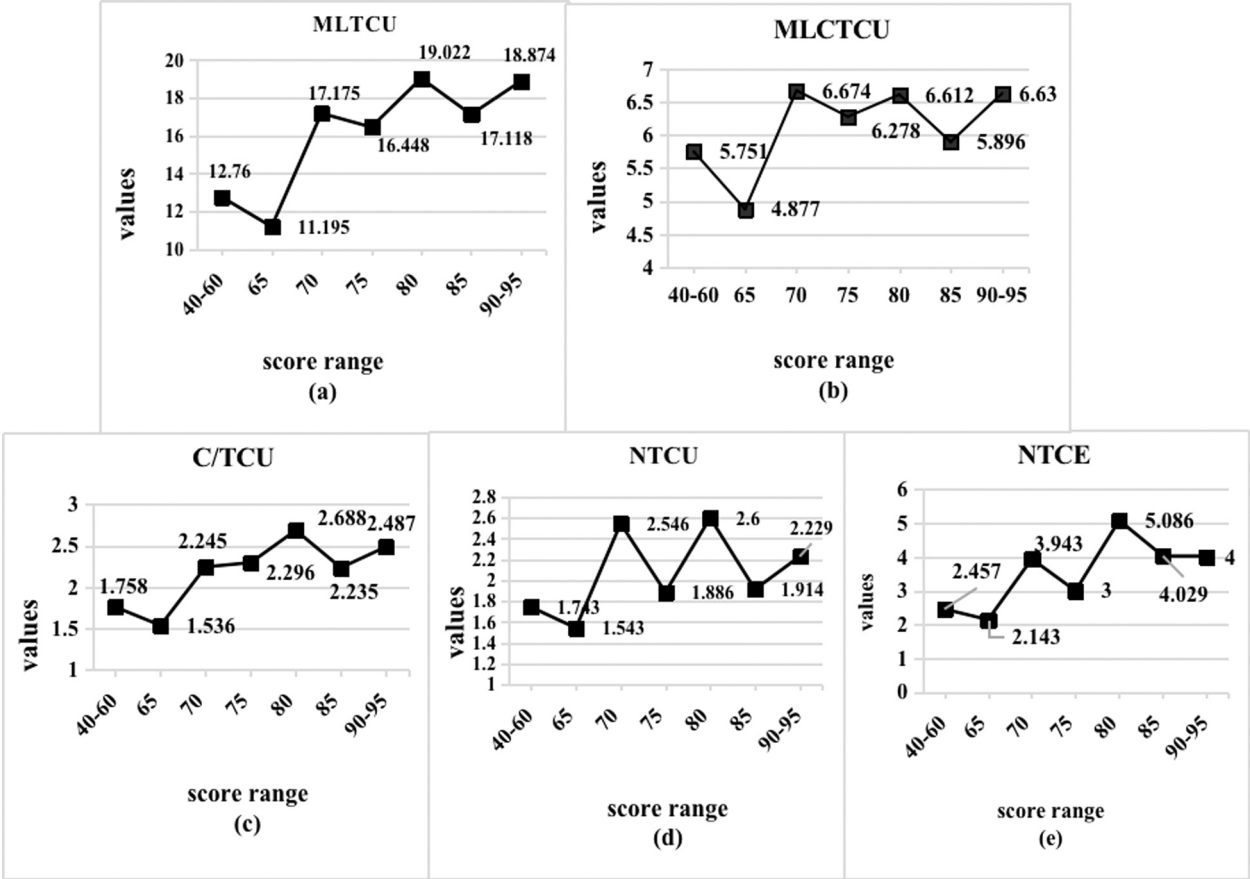


Fig. 3. Development trends of large-grained CSC indices across L2 Chinese writing scores.

Table 7

Summary of multiple regression model for large-grained SC indices.

Entry	Predictors included	R	R ²	Adjusted R ²	Std. Error	R ² Change
1	T/S	.343	.118	.114	11.141	.118
2	C/TCU	.381	.145	.138	10.988	.027

Table 8

Descriptive statistics of fine-grained complexity measures and their correlations with holistic score.

Category	Syntactic complexity indices	Mean (SD)	Correlation With Holistic Score
CSC indice	MSTTRDR-30	.422 (.031)	.497
	TCMDD	2.242 (1.020)	.257
	MDD	2.644 (.243)	.228
GSC indice	Adva_RATIO	.174 (.036)	.001
	Atr_RATIO	.144 (.037)	-.106
	Obj_RATIO	.125 (.028)	.055
	Subj_RATIO	.097 (.025)	-.252
	S_RATIO	.051 (.014)	-.322
	Cr_RATIO	.055 (.022)	.244
	Pobj_RATIO	.043 (.020)	.068
	Obja_RATIO	.027 (.017)	.084
	Coor_RATIO	.013 (.014)	-.151
	Sentobj_RATIO	.010 (.010)	-.111
	Epa_RATIO	.005 (.006)	.042
	Obj2_RATIO	.002 (.004)	-.188
	Csr_RATIO	.033 (.018)	.092
	Ma_RATIO	.0003 (.001)	-.159
	Qc_RATIO	.028 (.016)	-.007
CSC indice	Comp_RATIO	.020 (.013)	-.127
	Fc_RATIO	.020 (.013)	.078
	Ta_RATIO	.012 (.013)	.006
	Soc_RATIO	.010 (.010)	.223
	Auxr_RATIO	.007 (.008)	-.033
	Subobj_RATIO	.01 (.0105)	.236
	Oc_RATIO	.002 (.005)	-.205
	Plc_RATIO	.004 (.010)	-.069
	Va_RATIO	.005 (.008)	-.002
	Esa_RATIO	.006 (.008)	.191
	Ina_RATIO	.002 (.004)	.032
	Top_RATIO	.002 (.005)	-.041
	Dic_RATIO	.003 (.005)	.065
	Dfc_RATIO	.002 (.004)	-.018
	Dec_RATIO	.086 (.028)	-.067
	Beis_RATIO	.001 (.003)	.136
	Baobj_RATIO	.002 (.004)	.108

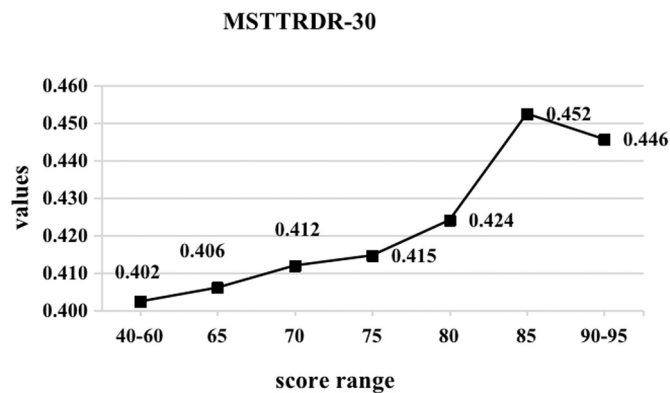
**Fig. 4.** Development trends of syntactic diversity MSTTRDR-30 across L2 Chinese writing scores.

Table 9
Summary of Multiple Regression Model for the Syntactic Diversity Index.

Entry	Predictors Included	R	R ²	Adjusted R ²	Std. Error	R ² Change
1	MSTTRDR-30	.497	.247	.244	10.253	.247

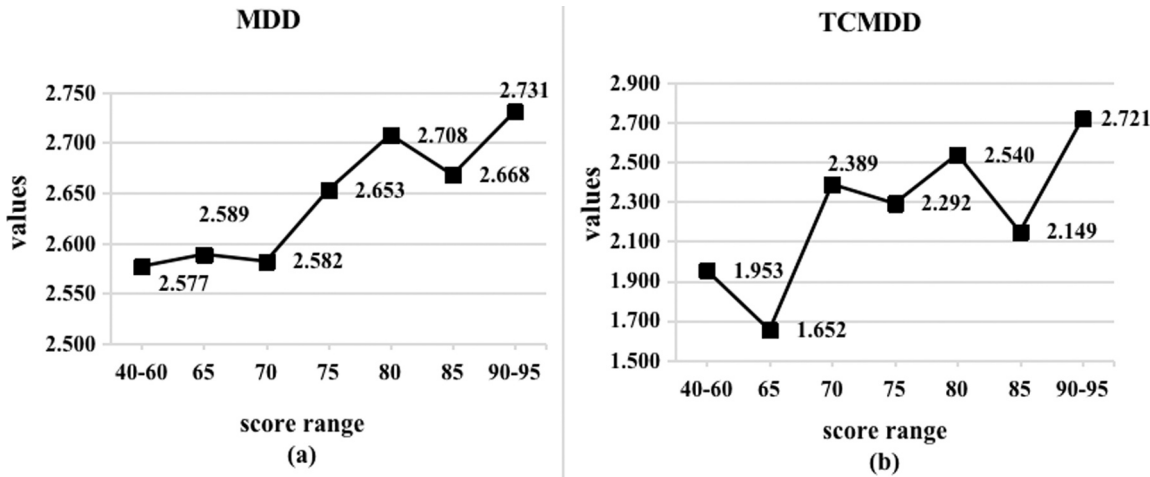


Fig. 5. Development trends of exemplary fine-grained indices across L2 Chinese writing scores.

are in the midst of acquiring typological features of Chinese that accentuate topics, undergoing a transition from a subject-prominent to a topic-prominent salience structure in their written work.

Figs. 6(b), 6(c), and 6(d) illustrate that the S_RATIO of ECSL learners decreased as their Chinese writing scores improved, while the Cr_RATIO and Csr_RATIO increased with higher scores. In the syntactic annotation system used in this study, *S* represents the main core predicate of a sentence, *Cr* indicates the connection between clauses in a complex sentence, and *Csr* represents the coordinating conjunctions that connect clauses within the corresponding complex sentence, such as in Example (1). In Example (1), "áoyè" (stay up late) dominates "Yīnwèi" (because), and the dependency relation is a conjunct relation (*csr*); "shuì lǎnjiào" (sleep in) dominates "suǒyǐ" (so), and the dependency relation is also a conjunct relation; the dependency relation between "shuì lǎnjiào" and "áoyè" is a complex sentence relation (*Cr*). The observed changes in these syntactic relations suggest that the sentence structures of ECSL learners become more complex, characterized by an increase in clauses and phrases, and a decrease in the number of main clauses. This indicates a potentially richer semantic content in their sentences. Additionally, statistical analysis shows that the average usage rate of coordinating conjunctions (3.31%) by ECSL learners is significantly higher than that of Chinese native speakers (1.35%), while the average usage rate of complex sentences (5.52%) is lower than that of Chinese native speakers (6.69%). Although the conjunction "Yīnwèi... suǒyǐ..." (because...so...) in Example (1) is a fixed collocation in Chinese complex sentences, Chinese native speakers tend to omit one or both of the conjunctions in the real language communication or writing. English is a hypotactic language with overt cohesion, where clauses are frequently arranged using connectives. In contrast, Chinese emphasizes parataxis and allows words, phrases, and clauses to be arranged sequentially without connectives, as long as the sentences maintain logical coherence (Nida, 1982). Therefore, this partly explains why ECSL learners tend to use more conjunctions. Jin (2007) also found that Chinese L2 learners with an English L1 background tend to use various conjunctions to connect clauses. Even at an advanced level, they believe that using explicit conjunctions can demonstrate the maturity of the Chinese language structure.

Example 1.

(a) Yīnwèi wǒ zuótiān áoyè le, suǒyǐ wǒ jīntiān zǎoshàng shuì lǎnjiào le.

因为我昨天熬夜了，所以我今天早上睡懒觉了。

Because I yesterday stayed up late (particle indicating completed action), so I today morning slept in.

"Because I stayed up late last night, I slept in this morning."

Regarding fine-grained CSC dependency relationship indices, Fig. 6(e) shows a decrease in the usage rate of the "De" (of) structure as a modifier as ECSL learners' Chinese writing scores improve. ECSL learners more frequently use the "De" structure as a complement (6.65%). Zhu (1982), a renowned Chinese grammarian, introduced the concepts of "adhesive structure" and "composite structure." For instance, in the attributive object structure, "mùtóu fángzi" (wooden house) represents the adhesive structure, while "mùtóu de fángzi" (house made of wood) represents the composite structure. In our selected Chinese dependency annotation system, the "De" structure complement represents the composite structure in the attributive object structure. From a linguistic modality perspective, there are

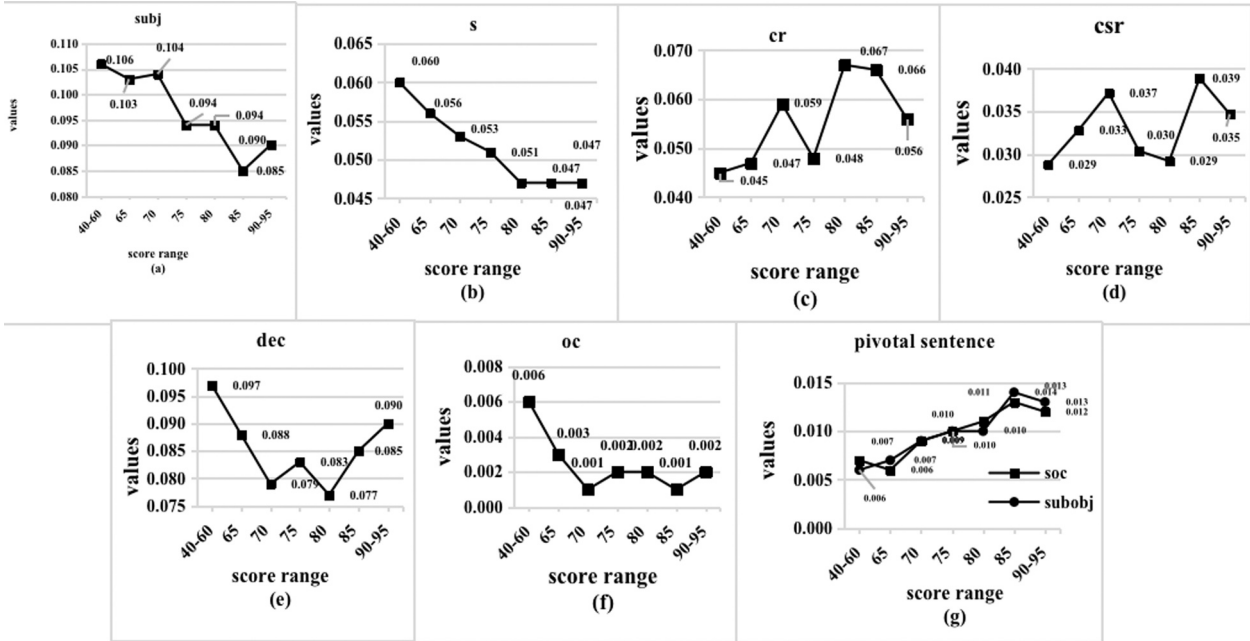


Fig. 6. Development trends of fine-grained dependency relationship ratio across L2 Chinese writing scores.

notable distinctions between modern Chinese adhesive and composite structures. Adhesive structures are generally considered more formal than composite structures with the "De" structure, especially in attributive object structures. However, in actual language usage, adhesive structures are predominantly employed in formal written modality, while composite structures with the "De" structure are more prevalent in informal spoken modality. Biber (1988) also believed that attributive adjectives can better package, integrate, and summarize information. Therefore, our results suggest that ECSL learners exhibit an oral tendency in their written language, reflecting lower syntactic complexity and integration. Additionally, according to Figs. 6(f) and 6(g), ECSL learners across L2 writing proficiency levels can gradually move from focusing and concentrating on typical Chinese syntactic structures to atypical unique syntactic structures. In regards to the Chinese language-specific SC indices, Oc represents ordinal grammatical expressions formed by grammatical devices, i.e., those formed by adding typical Chinese specialized morpheme markers (e.g., *dì*(第), *lǎo*(老) and *chū*(初)) to form Chinese ordinal expressions. As shown in Fig. 6(f), the Oc usage shows a downward trend, indicating a decrease in the ratio of ECSL learners using typically specialized marked ordinal syntactic expressions. The reduction in the use of typical syntactic structures reflects that ECSL learners gradually begin to incorporate non-typical types in the corresponding syntactic categories. To some extent, this reflects an increase in the syntactic diversity and complexity of the writing of ECSL learners. *Subobject* and *Subject Complement* together represent the pivotal construction, which is the special syntactic form in modern Chinese. In Fig. 6(g), as Chinese writing syntactic complexity improves, ECSL learners demonstrate an increased usage of pivotal constructions in their Chinese writing.

5.2.4. Predictive Power of Fine-Grained Syntactic Complexity Indices in Predicting Writing Scores

In order to address Question (3), we incorporated thirty-five syntactic complexity indices (comprising one diversity indice, two dependency distance indices, and thirty-two dependency relationship ratio indices) into the comprehensive model based on the syntactically annotated corpus. In the final stepwise regression model, nine syntactic complexity predictors are retained. As depicted in Table 10, this model includes nine significant predictors that account for 43.7% of the variance in holistic scores, making it the most effective predictor of writing scores in this study. The estimated standard error for the combined SC indices regression model is smaller than the standard deviation of the writing performance of ECSL learners (11.795). Additionally, the tolerance of the nine indices remains well above 0.5 after their inclusion in the regression model, and the VIF is less than 2. Consequently, there is no evidence of multicollinearity in the combination model of fine-grained SC indices. The P-P diagram and standardized residual histogram of the combination model both exhibit a normal distribution, and the regression coefficient is significant ($p < .05$), indicating a good fit for the regression model. A comparison of R^2 values shows that the combined model explains 29.2% more of the writing score variance than the large-grained SC indices model, and 19% more of the writing score variance than the syntactic diversity model. In this section, we have offered a solution to the third question that was previously raised in the preceding text.

We find that MSTTRDR-30 effectively measures Chinese L2 writing qualities. The increased variety of syntactic structures in the writings of ECSL learners is positively correlated with higher writing scores. The fine-grained combination model includes not only the generic syntactic structures common to both Chinese and English, but also the special syntactic structures in Chinese. This indicates that GSC and CSC indices complement each other in capturing the L2 syntactic complexity of language production. When evaluating the L2 writing qualities, L2 language evaluators should not only take into account the use of more complex syntactic structures, but also the basic syntactic structures employed in L2 writings.

Regarding common basic syntactic structures in Chinese and English, the frequency of attributive constructions can be seen as an indicator of the utilization of complex noun phrases. For example, the three complex noun phrase types mentioned in Lu and Wu (2022) are all marked with *Atr* in our annotation system. Thus, the presence of attributers in the combinatorial model indicates that the usage of noun phrases can effectively predict the writing performance of ECSL learners. The decrease in the usage of parallelism is partly indicative of a decrease in the number of syntactic structures that ECSL learners use at the same level and with equal constituents, thus also explaining the increase in their writing performance. In addition, the drop in parallelism is somewhat suggestive of a decrease in the number of syntactic structures utilized by ECSL students that have equivalent syntactic components, which could possibly account for the improvement in their writing qualities. Thus, the incorporation of *Atr_RATIO* in the combined model indicates that the utilization of noun phrases can function as a predictor for the writing proficiency of ECSL learners. ECSL learners who achieve higher scores in writing tend to utilize a range of sophisticated linguistic modifiers, thereby manifesting a heightened level of complexity in the syntactic constructions of their sentences. The decrease in the usage of *Coor* partly signifies a reduction in the number of syntactic structures that ECSL learners use at the same level with equal constituents, thereby contributing to the improvement in their writing performance.

In the realm of Chinese-specific syntactic structures, several indices are of particular significance in evaluating the quality of Chinese L2 writing. These include the ratio of predicate-complement, the ratio of adjunct of sentence end, the ratio of numeral adjunct, and the mean length of Topic-Comment-Unit dependency distance. Furthermore, our findings suggest that TCMDD, which measures the dependency distance within the topic chain, holds greater predictive power for the performance of ECSL learners compared to MDD. This demonstrates that ECSL learners not only have the capability to produce longer topic chains but that the syntactic complexity of these topic chains also escalates as their Chinese writing proficiency improves. This further implies that syntactic complexity indices incorporating Chinese linguistic typological features provide a more comprehensive explanation of Chinese L2 writing quality and better capture the evolving characteristics of interlanguage. These indices, tailored to the typological features of the target language, are better suited for reflecting these changes in interlanguage development. The predicate-complement structure in Chinese holds a unique position as an adverbial structure. According to the markedness differential hypothesis (Eckman, 1977), the marked nature of the predicate-complement structure makes it more challenging for Chinese L2 learners to acquire (Hao et al., 2021b). The inclusion of Comp ratios in the SC combination model emphasizes the importance of Chinese L2 learners' proficiency in handling complements, as it plays a significant role in assessing their Chinese writing quality.

Table 10
Summary of Multiple Regression Model for the combined fine-grained SC indices.

Entry	Predictors Included	R	R ²	Adjusted R ²	Std. Error	R ² Change
1	MSTTRDR-30	.497	.247	.244	10.253	.247
2	S_RATIO	.555	.308	.302	9.855	.060
3	Esa_RATIO	.576	.332	.324	9.700	.024
4	Atr_RATIO	.601	.361	.350	9.507	.029
5	Coor_RATIO	.619	.383	.370	9.360	.022
6	Sentobj_RATIO	.632	.399	.384	9.259	.016
7	Comp_RATIO	.644	.414	.397	9.161	.015
8	TCMDD	.652	.425	.405	9.095	.011
9	Ma_RATIO	.661	.437	.415	9.020	.012

6. Conclusions and implications

Based on a syntactically annotated Chinese L2 writing corpus, we investigated the relationship between syntactic complexity and writing quality of ECSL learners. We utilized a range of syntactic complexity indices at both the large and fine levels. Our analysis revealed that only T/S and C/TCU were included in the regression model, accounting for a mere 14.5% of the variance in Chinese writing scores. To address Question (4), we present the following summary. Among the large-grained indices, the GSC indice that effectively predicts the performance of Chinese L2 writing is the number of T-units per sentence, while the CSC indice that effectively predicts the performance of Chinese L2 writing is the number of Clauses per topic chain unit. In terms of fine-grained syntactic annotated complexity indices, the syntactic diversity indice alone can explain 24.7% of the variance in writing scores. However, the combined model of fine-grained syntactic complexity and syntactic diversity indices, based on the syntactically annotated corpus, can explain 43.7% of the variance in holistic scores, demonstrating the strongest predictive power for Chinese L2 writing qualities in this study. This indicates that fine-grained syntactic complexity indices provide a more comprehensive explanation of L2 writing qualities. Specifically, the fine-grained GSC indices that have been found to effectively predict Chinese L2 writing qualities are S_RATIO, Atr_RATIO, Coor_RATIO and Sentobj_RATIO. Furthermore, Chinese-specific indices, including Esa_RATIO, Comp_RATIO, Ma_RATIO and the dependency distance of topic chains, have also been determined to effectively predict Chinese L2 writing performance.

The research findings presented above have significant implications both theoretically and practically. The large-grained and fine-grained indices considered in the current study exhibit varying performances in predicting the quality of L2 Chinese writing. The more effective prediction of writing quality by fine-grained indices supports the notion that fine-grained indices empower researchers to identify specific syntactic constructions that appear to evolve at each developmental level (Kyle & Crossley, 2018). Our study results demonstrate that the combined use of GSC and CSC indices provides a proficient prediction of L2 writing proficiency. Therefore, it is advisable to employ multidimensional syntactic complexity indices rather than a selective approach to conceptualize syntactic complexity in predicting the quality of second language Chinese writing (Lu & Wu, 2022; Norris & Ortega, 2009). Meanwhile, educators must avoid an exclusive emphasis on certain specialized structures of the target language in the teaching process. Simultaneously, there should be a reinforced focus on the usage of basic structures by L2 learners, ensuring a comprehensive and integrated approach to instruction. Syntactic diversity holds considerable potential in predicting the quality of L2 writing. Furthermore, the concurrent application of both syntactic elaboration and diversity indices yields a more holistic portrayal and understanding of syntactic complexity in its entirety. This supports and aligns with the perspective proposed by De Clercq and Housen (2017). This also implies that L2 teachers should explicitly provide guidance to L2 learners on how to generate a greater variety of syntactic structures. Furthermore, our findings indicate that typological features inherent in the English native language, such as subject prominence and inflectional morphology, influence the syntactic complexity of ECSL learners. With regard to this particular discovery, Chinese teachers should guide ECSL learners to observe the structural similarities and differences between their native language and the target language (Chinese). When creating Chinese instructional materials, it is also essential to consider the native language factors of L2 learners and strive to develop more culturally tailored materials. Furthermore, our research data can be leveraged for machine learning or deep learning, thus enabling the effective development of an automated scoring system for Chinese L2 writing. For instance, we may select linguistic features that exhibit a correlation with manually assigned scores. By utilizing machine learning algorithms (such as support vector machines, random forests, and neural networks) and identifying critical features, a more potent predictive model can be established (Mizumoto & Eguchi, 2023). This model has the potential to aid second language educators in more effectively understanding the quality of writing and the development of L2 learners (Crossley, 2020).

While we have obtained some significant findings in this study, we should also acknowledge the limitations of this research article. First, our data only encompass essays written by L2 learners from one native language background (English) and within one genre (narrative essays). Consequently, it is challenging to ascertain whether these findings are applicable to writing in other genres or for learners from different linguistic backgrounds, as both the task genre (Qin & Uccelli, 2016) and learners' native language backgrounds (Lu & Ai, 2015) can impact syntactic complexity in L2 writing. Therefore, in the future, we should expand our research to investigate the relationship between syntactic complexity indices in essays of various genres written by Chinese L2 learners from diverse native language backgrounds and the quality of their writings. Second, as pointed out by one of the reviewers, the analysis in the study is conducted across score levels rather than across proficiency levels. Therefore, future research on the relationship between Chinese L2 writing quality and syntactic complexity indices should pay attention to longitudinal changes over time. For instance, future research could longitudinally track the syntactic complexity of writings produced by L2 learners, considering individual learners, over an

extended period. Third, in terms of syntactic diversity, we have employed only the MSTTRDR-30 as a syntactic diversity indice for predicting the quality of Chinese L2 writing in this article. In the future, we will explore additional syntactic diversity indices that are suitable for assessing the quality of Chinese L2 writing.

CRedit authorship contribution statement

Xuelin Wang: Conceptualization, Data curation, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Shuai Bin:** Software. **Qihao Yang:** Software. **Haitao Liu:** Conceptualization, Funding acquisition, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Yuxin Hao:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing.

Data Availability

Data will be made available on request.

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Yuxin Hao is a professor of linguistics and applied linguistics at Huaqiao University. His research interests include second language acquisition and quantitative linguistics. He is the author of more than 50 scientific publications about applied linguistics.

Xuelin Wang is an academic doctoral candidate in linguistics and applied linguistics at Jinan University. Her research interest is second language acquisition, language stylistics and quantitative linguistics.

Shuai Bin is an academic doctoral candidate at School of Chinese as a Second Language, Peking University, China. His research interests include corpus linguistics, second language acquisition and computational linguistics.

Qihao Yang is an academic master candidate in natural language processing at South China Normal University. His research interests include computational linguistics and multimodal deep learning.

Haitao Liu is a Qishi distinguished professor of linguistics and applied linguistics at Zhejiang University. His research interests include dependency syntax, quantitative linguistics and applied linguistics. He is the author of more than 200 scientific publications about language and linguistics.